MINIGUN

Minigun Drum Sampler

User Manual · VST3 & Standalone · version 0.3

16 pads · velocity layers · round-robin / random · portable kits · 17 stereo outputs
Windows 10/11, 64-bit. Built with JUCE 8.

1. What Minigun is

Minigun is a lightweight drum sampler in the spirit of hardware pad machines. Sixteen pads each hold one instrument: a MIDI note, up to eight velocity layers, and any number of samples per layer that alternate round-robin or at random. Every pad has its own volume, pan, pitch, envelope, choke group and output bus, so a whole kit can be mixed inside your DAW on separate tracks.

Kits are **portable**: when you save a kit, Minigun copies every sample into the kit folder and stores relative paths, so the folder can be moved to another project or another computer without broken links.

Installation

- **VST3**: copy `Minigun.vst3` (the whole folder) to `C:\Program Files\Common Files\VST3\`, then rescan plug-ins in your DAW. Minigun appears as *VST3i: Minigun (Dallas Audio)*.
- **Standalone**: run `Minigun.exe`. Use *Options → Audio/MIDI settings* to pick an audio device and a MIDI input. Only the Main output is available in Standalone.

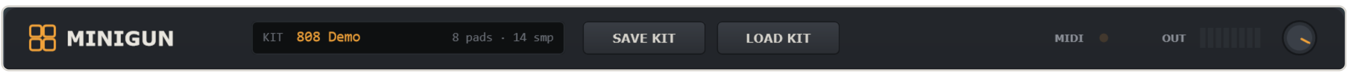
> The Standalone version and a DAW that holds an exclusive ASIO driver cannot use the same audio device at the same time. If Standalone shows *0 voices* and no meter movement, close the DAW or choose another device.

2. Interface overview

The fixed 1200 × 720 window. Everything is always visible; nothing is hidden in menus.

1. **Header** — kit name, Save / Load kit, MIDI indicator, output meter, master volume.
2. **Pads** — 16 pads in MPC order (pad 1 bottom-left, pad 13 top-left). Click to play, drop audio files to assign.
3. **Pad editor** — every setting of the selected pad: name, note, choke, output, mix and envelope knobs.
4. **Velocity layers** — the velocity map of the selected pad and the samples inside each layer.
5. **Browser** — a file browser with instant preview; drag files onto pads or layers.
6. **Footer** — kit folder, active voices, and how many output buses the host has enabled.

3. Header



Control	What it does
KIT display	Shows the kit name, how many pads have samples and the total sample count. The name is taken from the folder you save into (a kit called *Untitled Kit* is renamed automatically on first save).
SAVE KIT	Saves the kit. The first time it opens a folder picker (default location: `Documents\Minigun Kits`). Choose or create a folder: Minigun writes `kit.json` there and **copies every sample** into a `samples\` sub-folder. If the kit already has a folder, it saves there directly without asking.
LOAD KIT	Opens a folder picker. Select a folder that contains `kit.json`; all pads, layers and settings are replaced by that kit.
MIDI	Lights amber for a moment whenever a MIDI note-on arrives. Use it to confirm the DAW is routing MIDI to the plug-in.
OUT	Eight-segment stereo peak meter of the plug-in output (the loudest of all enabled buses). Teal segments are healthy, amber is hot.
Master knob	Master volume, −60 … +6 dB. This is the only DAW-automatable parameter; everything else belongs to the kit.

4. Pads

Pad 1 is selected (teal outline). Loaded pads show their name and one LED per velocity layer.

What a pad shows

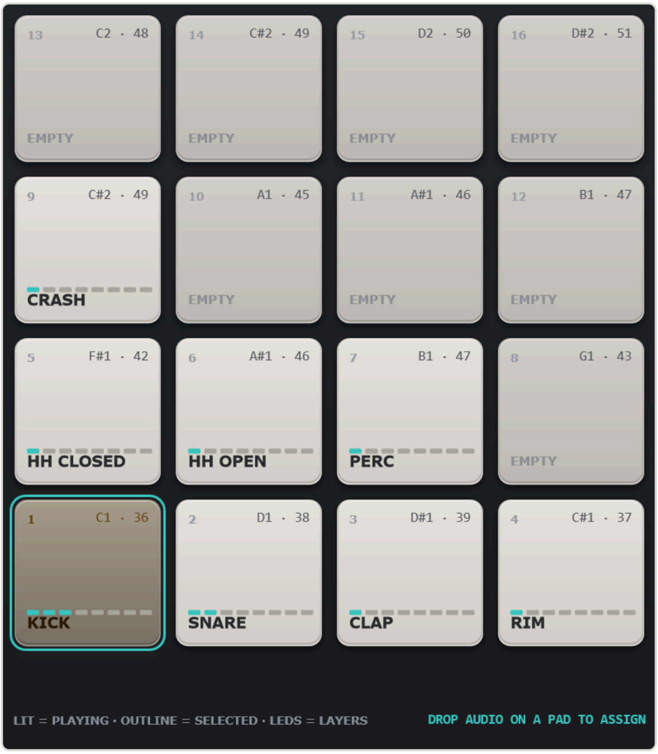
- **Number** (top-left) — pad 1…16 in MPC order.
- **Note** (top-right) — the MIDI note that triggers the pad, e.g. `C1 · 36`.
- **LED strip** — one lit LED per velocity layer (up to 8).
- **Name** — the pad name, or *EMPTY*.

Pad states

- **Empty** — darker rubber, no LEDs.
- **Loaded** — light rubber, LEDs show the layer count.
- **Selected** — teal outline; the editor and layer panel show this pad.
- **Playing** — the pad lights up in the colour of the velocity layer that played (see §6).

Interactions

- **Click** a pad to select it and play it. The velocity follows the click height: top edge = 127, bottom edge = 1.
- **Drop audio files** from Explorer or from the Browser onto a pad. On an empty pad they become layer 1 (velocity 1–127). On a loaded pad they become a new top layer that takes the upper half of the previous top layer's range. Several files dropped together go into the same layer.
- A pad is named automatically after its first sample when it has no name yet.



A soft hit (layer 1) lights the Kick pad dark brown...

...a hard hit (layer 3) lights it bright amber. The colours match the layer bar in §6.

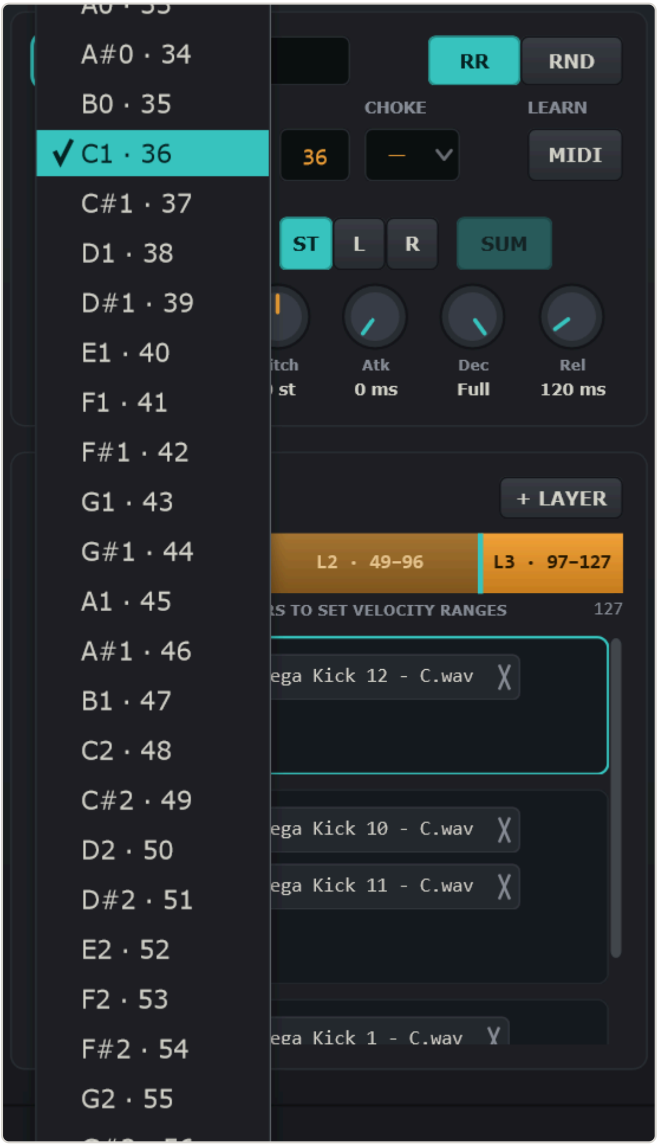
5. Pad editor

The editor always shows the selected pad (badge *1* = pad 1).

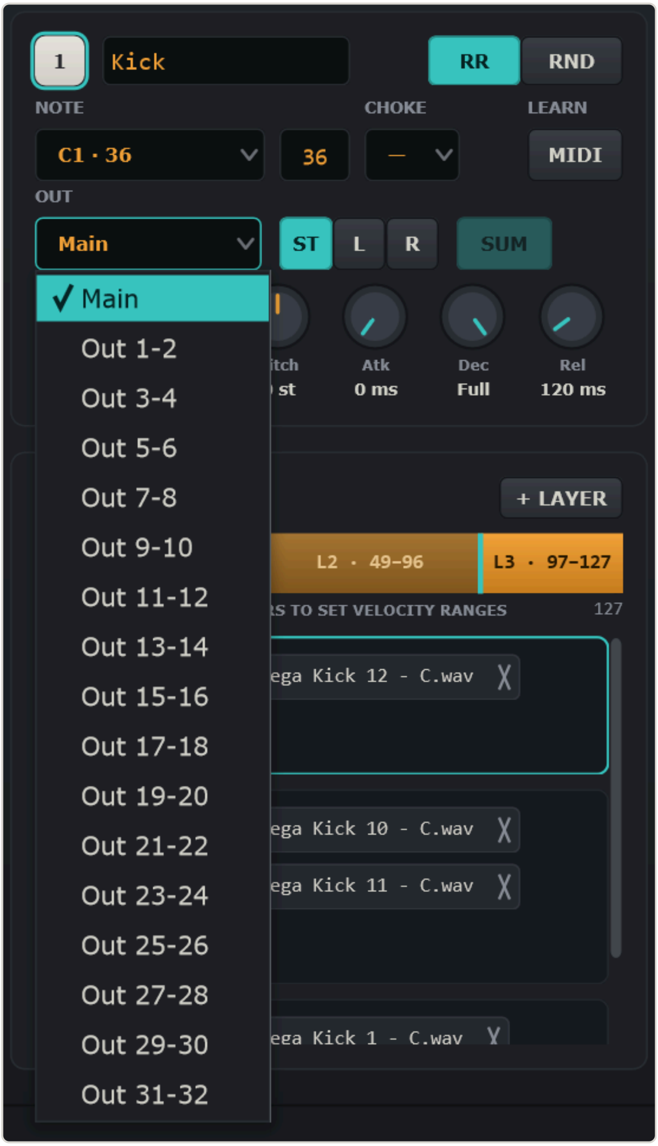
Control	What it does
Badge + name	The pad number and an editable name. Click the name, type, press `Enter`. The name is shown on the pad.
RR / RND	How samples inside a layer alternate. **RR** (round-robin) plays them in order 1, 2, 3, 1...; **RND** picks one at random on every hit. Use several similar samples per layer for a natural, non-machine-gun feel.
NOTE	The MIDI note that triggers the pad. Pick it from the drop-down (all 128 notes, `C1 · 36` style) or type the number into the field next to it and press `Enter`. The mouse wheel over the drop-down steps by one semitone. Several pads may share a note: they all play together.
CHOKE	Choke group, — or 1...8. Pads in the same group cut each other off: when one is hit, the others in that group are released immediately. Classic use: closed and open hi-hat in group 1, so the closed hat stops the open one.
LEARN / MIDI	MIDI learn. Click the button (it lights teal), then hit a key or pad on your controller: that note is assigned to the selected pad.
OUT	Output bus of the pad: **Main** or one of the 16 stereo pairs **Out 1-2 ... Out 31-32**. See §9 for routing in the DAW.
ST / L / R	Stereo or mono output. **ST** sends the pad to both channels of the pair with panning. **L** sends a mono signal to the first (odd) channel only, **R** to the second (even) channel. In mono mode the OUT list shows single channels (*Main L*, *Out 3*, ...) and Pan is ignored.
SUM	Only for L/R modes. On: a stereo sample is summed to mono ((L+R)/2) before it goes to the single channel. Off: only the sample channel that matches the side is used. Mono samples are unaffected.
Vol	Pad volume, −60 ... +12 dB.
Pan	Stereo position, L100 ... C ... R100 (constant power). Not used in mono modes.
Pitch	Transposition by resampling, ±12 semitones in 0.1 steps. Positive values are shorter and brighter.
Atk	Attack, 0 ... 500 ms: fade-in from the start of the sample. 0 ms = the natural transient.
Dec	Decay / hold length, 1 ... 5000 ms. Fully right = **Full**: the sample plays to its end. Shorter values cut the tail and start the release.
Rel	Release, 1 ... 2000 ms: the fade-out applied when the decay time ends, or when the pad is choked by another pad of its group.

Snare pad routed to the stereo pair *Out 1-2*.

Closed hi-hat: choke group 1, panned right, *Out 5-6*, short 60 ms release.

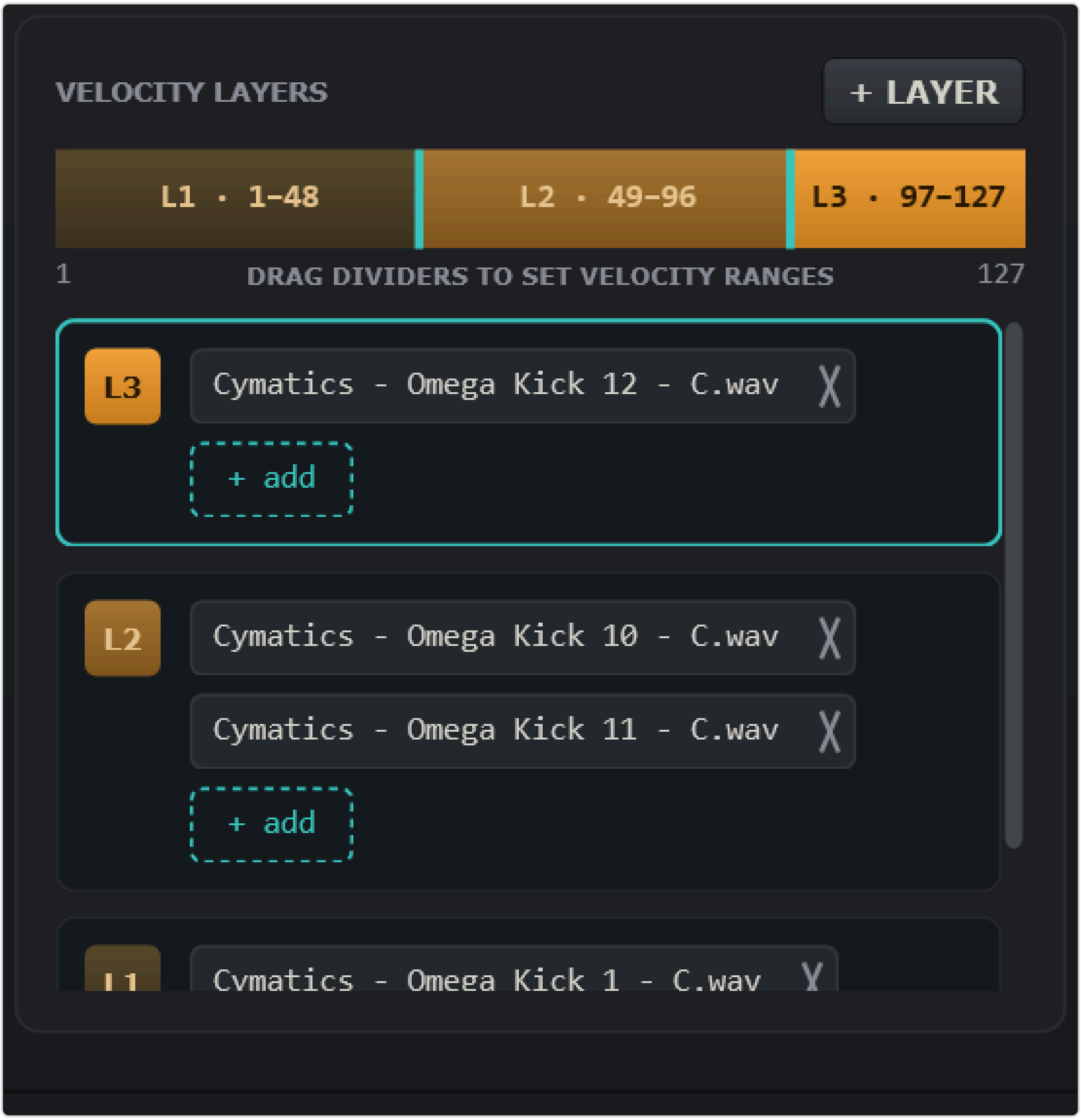


The NOTE drop-down lists all 128 notes with their numbers.



The OUT drop-down: Main plus 16 stereo pairs.

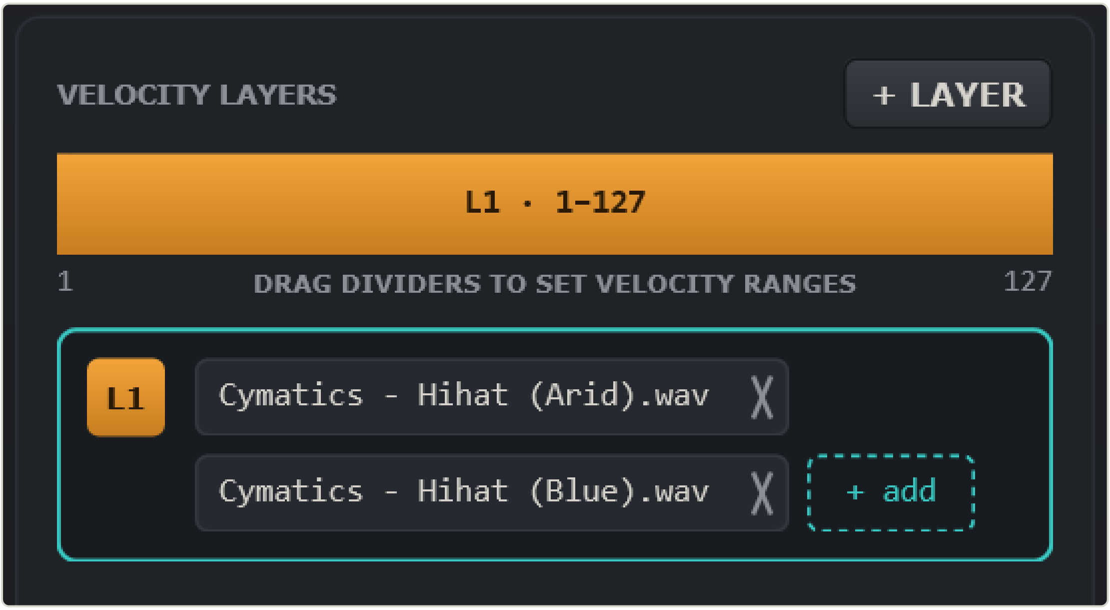
6. Velocity layers



Three layers on the Kick pad. Layer 2 holds two samples that alternate round-robin.

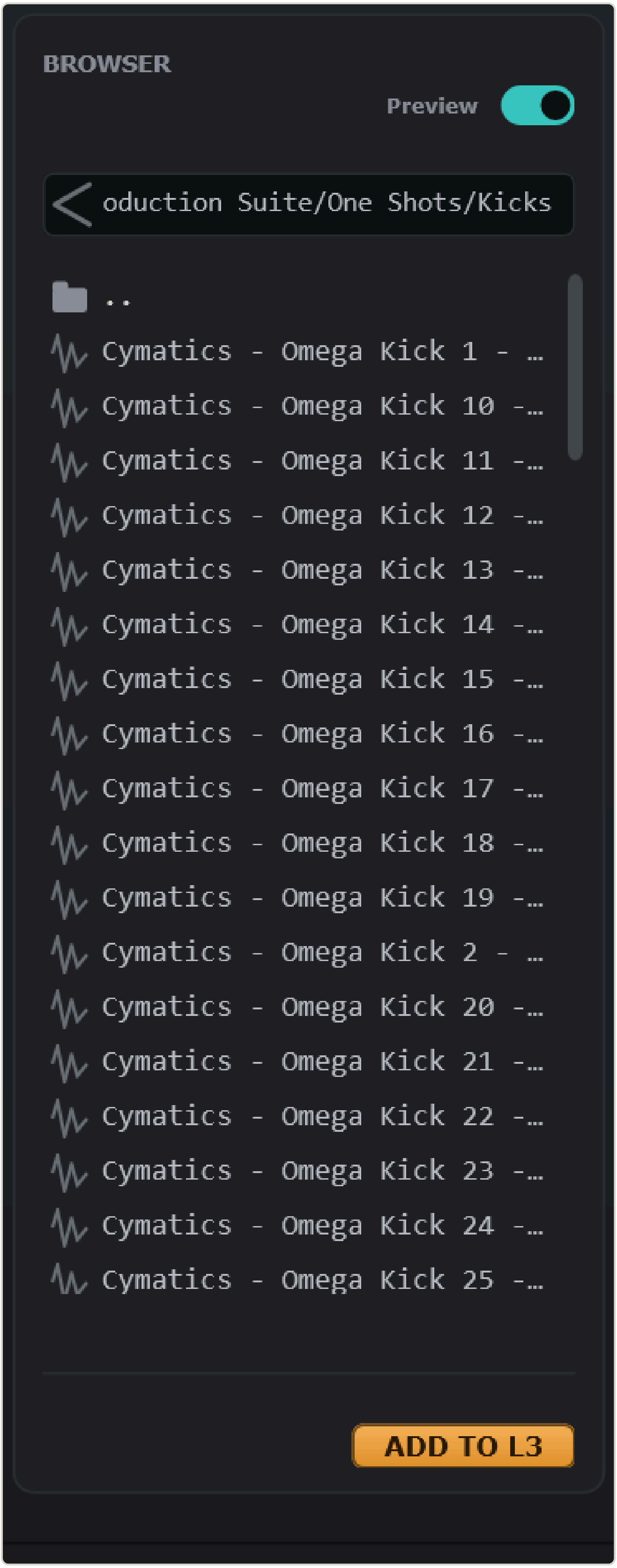
A velocity layer is a velocity range (1–127) with its own list of samples. When a note arrives, Minigun finds the layer whose range contains the velocity, then picks one of its samples by the pad's RR/RND rule. Layers are contiguous and never overlap; the panel keeps them valid for you.

Control	What it does
Range bar	One coloured segment per layer, from dark (soft) to bright amber (hard). **Drag the teal dividers** to move the boundaries. **Click inside a segment** to audition that layer: the pad plays at the clicked velocity, so the layer and the RR/RND rule apply exactly as with a MIDI note.
+ LAYER	Adds a layer on top: the current top layer is split in half and the new (empty) layer takes the upper part. Up to 8 layers.
Layer rows	One row per layer, top layer first. The coloured badge (*L1*, *L2*...) matches the bar. Click a row to make it the **selected layer** (teal frame): the Browser's *ADD TO Lx* button and double-clicks add files to it.
Sample chip	One chip per sample. **Click** the name to preview the sample. **Click ×** to remove it from the layer (the file on disk is not touched).
+ add	Opens a file dialog (multi-select) and adds the chosen files to that layer.
Right-click a row	*Delete layer*: removes the layer and gives its range to the neighbour below (or above for the lowest layer).
Drop target	Files dragged from Explorer or the Browser can be dropped on a row or on a bar segment (that layer) or anywhere else in the panel (the selected layer). The target is highlighted with a dashed teal frame while you drag.



A single full-range layer with two hi-hat samples: every hit alternates between them.

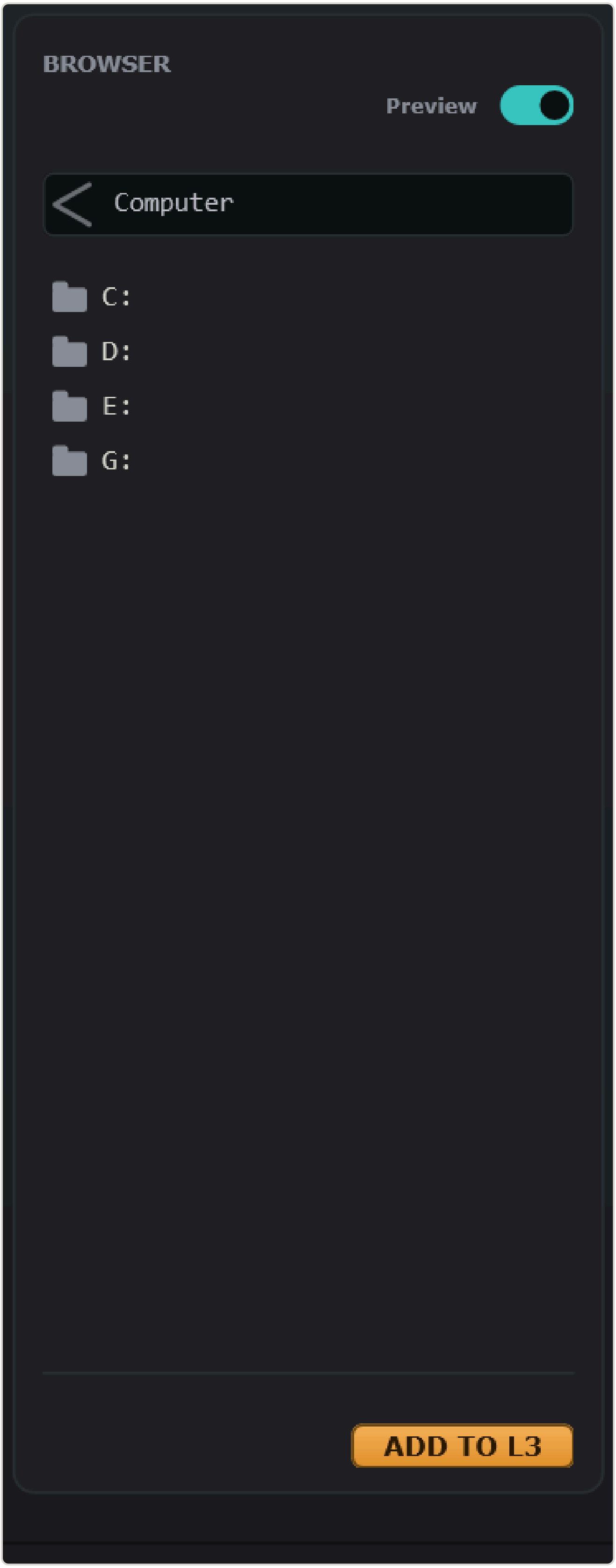
7. Browser



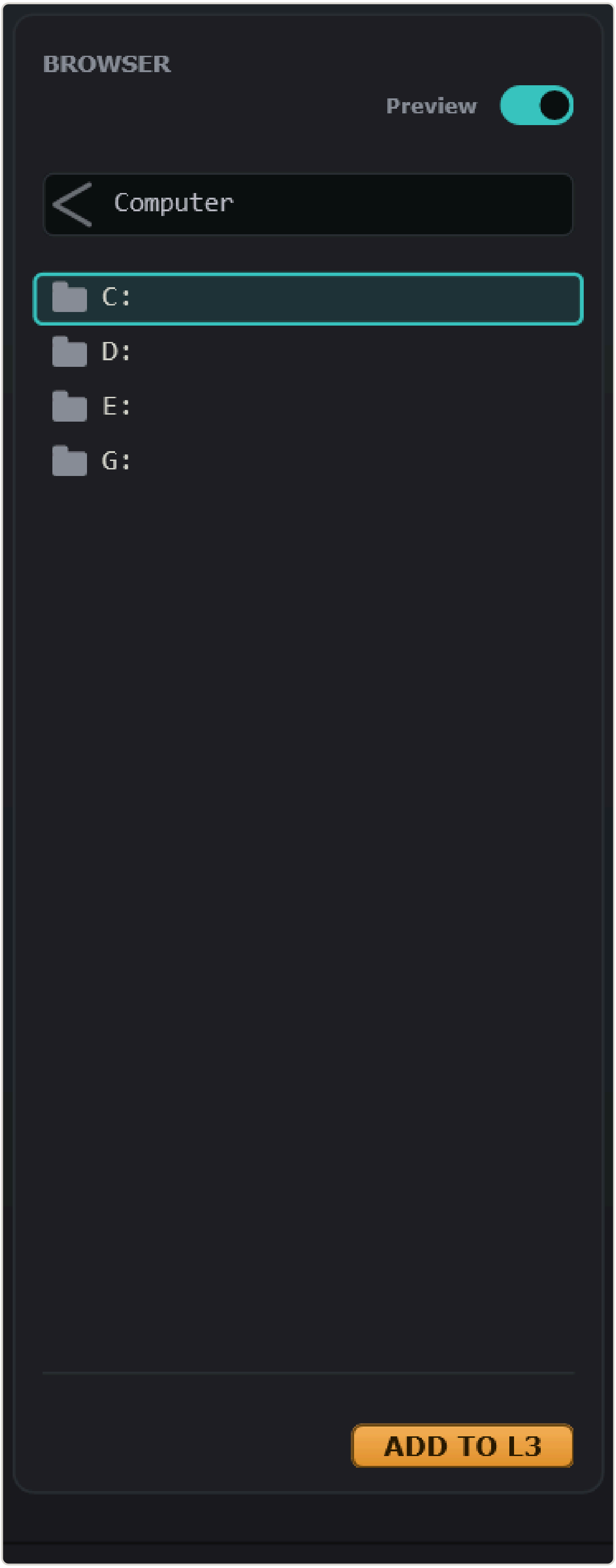
Audio files show a waveform icon; the playing file shows a play triangle. The footer shows the waveform and format of the selected file.

Control	What it does
Preview	When on, a single click on a file plays it through the plug-in's Main output. Turn it off to browse silently.
‹ (arrow)	Go up one folder. From the root of a drive it opens the *Computer* level with all drives.
Path field	Shows the current folder. Click it, type a path to a folder or to an audio file and press `Enter`. A file path opens its folder and selects the file. `Esc` reverts.
Folder row	Single click frames the folder; double-click or `Enter` opens it.
File row	Single click selects and previews. Double-click or `Enter` adds the file to the selected layer of the selected pad. Drag a row onto a pad or onto the layer panel to add it there.
Waveform + info	Overview of the selected file with sample rate, bit depth and length.
ADD TO Lx	Adds the selected file to layer x of the selected pad (x = the layer selected in §6).

Supported formats: WAV, AIFF, FLAC, OGG, MP3, WMA. The last folder is remembered between sessions.



The *Computer* level lists all drives.



A clicked folder gets a teal frame; double-click to enter.

Navigation shortcuts

Action	Result
`Backspace`	Up one level.
Mouse Back / Forward buttons	Back / forward through the folder history (while the pointer is over the browser).
`Alt` + `←` / `Alt` + `→`	Back / forward through the history.
`Enter`	Open the framed folder or add the selected file.
`↑` / `↓`	Move through the list.

8. Kits: saving, loading, moving

A kit is a folder. **SAVE KIT** creates this structure:

```
My Kit\
  kit.json          all pad settings, layers and sample references
  samples\
    kick_01.wav     every sample used by the kit, copied here
    kick_02.wav
    ...
```

- Sample paths inside `kit.json` are relative to the kit folder, so the folder can be zipped, moved to another disk or another computer, and loaded with **LOAD KIT** without any missing files.
- Files that are already inside `samples\` are not copied again. If two different files share a name, the second gets a `_2` suffix.
- Removing a chip never deletes a file from disk.
- In a DAW the complete kit is also stored in the project, so a project reopens exactly as it was saved even if you never pressed SAVE KIT. Save a kit when you want to reuse it in another project.

> If a sample cannot be found when a kit or project is loaded, the pad keeps its settings but that sample is silent. Add the file again from the Browser or replace it via + *add*.

9. Routing pads to separate DAW tracks

Minigun exposes **17 stereo outputs**: Main and Out 1-2 ... Out 31-32 (34 channels). Each pad chooses its bus with the OUT drop-down, and can send a mono signal to a single channel with the L / R modes.

REAPER example

1. Insert Minigun on a track. The FX window shows a button like *2/34 out*.
2. Set the track's channel count (track routing button) to as many channels as you need, e.g. 8 for Main + Out 1-2 + Out 3-4 + Out 5-6. REAPER enables exactly that many plug-in buses; the footer shows *outs 4/17*.
3. In Minigun set the Snare pad to *Out 1-2*, the Clap to *Out 3-4*, the hats to *Out 5-6*.
4. Create receive tracks: on a new track add a receive from the Minigun track with source channels 3/4 for the snare, 5/6 for the clap, 7/8 for the hats. Or use *Build multichannel routing for this plug-in* in the plug-in I/O window.

Other DAWs work the same way: enable the plug-in's aux outputs and route each pair to its own track. A pad set to an output the host has not enabled falls back to Main, so nothing ever goes missing.

Mono outputs

To send a pad to one channel only, choose the pair in OUT and set **L** (odd channel) or **R** (even channel). With **SUM** on, a stereo sample is folded to mono; with SUM off, only the matching side of the sample is used.

10. Step-by-step examples

A kick with three velocity layers

1. Click pad 1. In the Browser go to your kick samples.
2. Double-click the softest sample: it becomes layer 1 (1–127).
3. Press **+ LAYER**: layer 2 appears above (65–127). Double-click a medium sample to add it there, then a second medium sample for round-robin.
4. Press **+ LAYER** again and add the hardest sample to layer 3.
5. Drag the dividers so the ranges feel right, e.g. 1–48, 49–96, 97–127. Click inside each segment to audition it.

Closed and open hi-hat that choke each other

1. Drop a closed hat on pad 5 and an open hat on pad 6.
2. Set **CHOKE** to *1* on both pads.
3. Give the closed hat a short **Rel** (about 60 ms) so it cuts the open hat cleanly.

A snare on its own mixer track

1. Select the snare pad, set **OUT** to *Out 1-2*.
2. In the DAW enable the plug-in's second output pair and route channels 3/4 to a new track.

A mono percussion hit on channel 5

1. Select the pad, set **OUT** to *Out 5-6*, press **L**: the list now reads *Out 5*.
2. Leave **SUM** on if the sample is stereo and you want both sides folded in.

Assign a controller pad

1. Select the pad, press **MIDI** (LEARN).
2. Hit the pad on your controller. The note is shown in NOTE and on the pad.

11. Shortcuts and gestures

Where	Gesture	Result
Pad	Click (top / bottom)	Play loud / soft, select the pad
Pad	Drop files	New layer with those files
Range bar	Click segment	Audition that layer (RR/RND applies)
Range bar	Drag divider	Change velocity ranges
Layer row	Right-click	Delete layer
Chip	Click / click ×	Preview / remove sample
NOTE drop-down	Mouse wheel	±1 semitone
Browser	`Backspace`	Up one folder
Browser	Mouse Back / Forward, `Alt` + `←` / `→`	History back / forward
Browser	Double-click / `Enter`	Open folder or add file to the selected layer
Path field	Type path + `Enter`	Jump to folder or file

12. Footer and troubleshooting

- **Kit folder** — where the kit was saved or loaded from; *(unsaved)* until the first SAVE KIT.
- **voices** — samples currently sounding (32 maximum; the oldest is stolen when exceeded).
- **outs N/17** — how many output buses the host has enabled. In Standalone this is 1/17; in a DAW it grows with the track's channel count.

Symptom	Check
No sound in Standalone	Options → Audio/MIDI settings: pick an output device. Another program (a DAW with ASIO) may hold the device exclusively.
Pads light but a pad is silent	Its samples may be missing (moved or renamed on disk). Re-add them; then SAVE KIT so they are copied into the kit folder.
Pad routed to Out 5-6 comes out of Main	The host has not enabled that bus. Increase the track channel count or enable the plug-in's outputs; the footer *outs* value shows how many are active.
Plug-in not listed in the DAW	Make sure `Minigun.vst3` is in `C:\Program Files\Common Files\VST3` and rescan plug-ins.
Wrong pad plays from the keyboard	Check NOTE on each pad; several pads may share a note. Use LEARN to assign quickly.